

- 3 (a) gravitational force = $m_p g = 1.67 \times 10^{-27} \times 9.81$
 $= 1.6 \times 10^{-26} \text{ N}$ M1
- electric force = $q_p E = q_p (V/d) = 1.6 \times 10^{-19} \times 270 / (1.8 \times 10^{-2})$ M1
 $= 2.4 \times 10^{-15} \text{ N}$
- electric force is very much greater than gravitational force A1
- (b) (i) either both potentials are positive or same sign $\rightarrow$ charges same sign M1
 A1
 or gradients of graph are positive & negative so E fields in opposite directions so same sign
- (ii) $x = 11 \text{ nm}$ (allow value of x between 10 nm and 13 nm) B1

(iii) $V_A = 0.36 \text{ V}$ when $x = 8 \text{ nm}$ (any value from graph of V_A)

$$V_A = \frac{Q_A}{4\pi\epsilon_0 x}$$

$$Q_A = 3.2 \times 10^{-19} \text{ C}$$

M1

A1

(iv) $V_P = 0.14 + 0.29 = 0.43 \text{ V}$ (allow $0.42 \text{ V} \rightarrow 0.44 \text{ V}$)

$$KE_i + PE_i = KE_f + PE_f$$

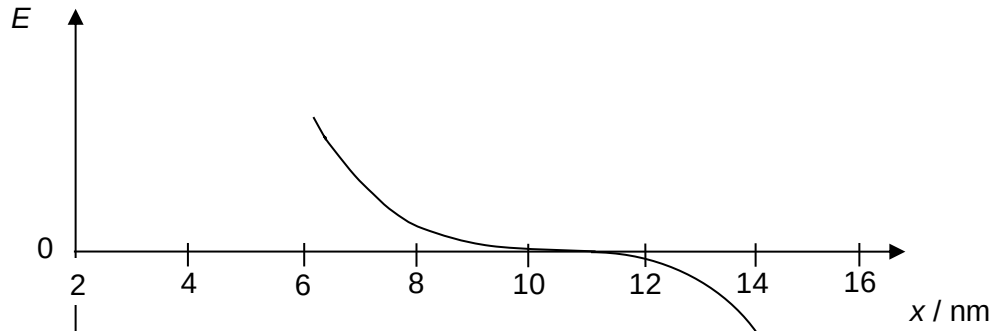
$$E_K + 0 = 0 + (1.60 \times 10^{-19})(0.43)$$

$$E_K = 6.88 \times 10^{-20} \text{ J}$$

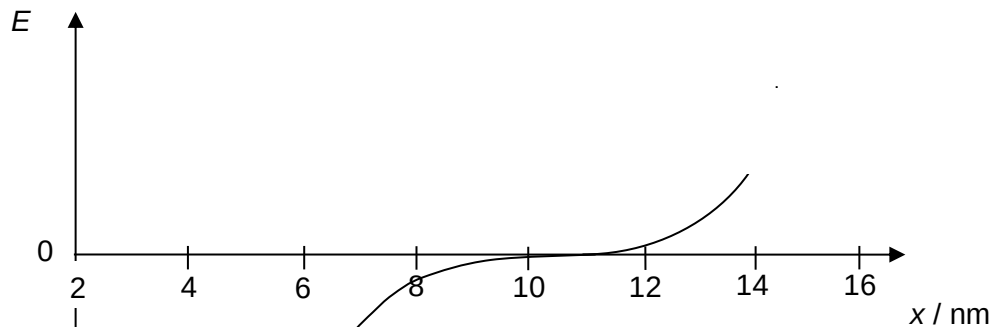
M1

A1

(v)



Or



E fields in opposite directions; correct curvature

B1

Graph crosses the x-axis at $x = 11 \text{ nm}$ (same as ans given in b(iii); allow $\pm 0.5 \text{ nm}$ for point plotted) and magnitude of E at $6 \text{ nm} > E$ at 14 nm with correct range (6 nm to 14 nm)

B1